Bola Sameh Dawoud

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LinkedIn | GitHub | Portfolio

Education

Bachelor of Mechatronics Engineering (Sep. 2022 – Jun. 2027)

Faculty of Engineering, Ain Shams University

Senior Mechatronics Engineering Student || Cumulative GPA: 3.1/4 (Very good)

Work Experience & Internships

Freelance Mechatronics Engineer | Self-Employed (Feb. 2026 – Present)

- Designed custom CAD models and PCB layouts for hardware prototyping projects.
- Developed and deployed machine learning solutions for client automation use cases.

Mechatronics Engineer | Lotus-Power (Jun. 2026 – Sep. 2026)

- Performed research and technical analysis for beekeeping and farm management devices to support product design and development.
- Worked with embedded systems, and software teams on IoT systems, PCB design, electronic circuits, manufacturing, and technical documentation.

Workshop Maintenance Engineer Intern | SMG Engineering Automotive Company (Aug. 2026 – Aug. 2026)

- Executed preventive and corrective maintenance on specialized workshop tools and heavy industrial equipment, optimizing machinery performance and minimizing operational downtime.
- Conducted technical diagnostics, troubleshooting, and calibration of automotive service equipment in strict adherence to industry safety standards and manufacturer specifications.

Workshop Engineer Intern | SNA Mercedes-Benz (Jun. 2026 – Jul. 2026)

- Performed routine vehicle service and preventive maintenance on Mercedes-Benz passenger cars, including brakes, suspension, coolant, and tires.
- Conducted basic electrical and mechanical diagnostics on engine and vehicle systems using standard workshop tools and manufacturer procedures.

ML Engineering Trainee | Digital Egypt Pioneers Initiative (DEPI) (Nov. 2025 – Jul. 2026)

- Deployed Deep Learning and Machine Learning models for industrial Predictive Maintenance and time-series anomaly detection.
- Integrated Computer Vision (OpenCV) and NLP pipelines into Robotics frameworks for autonomous navigation.

IoT Engineering Trainee | Samsung Innovation Campus (SIC) (Aug. 2025 - Dec. 2025)

- Developed and debugged end-to-end IoT applications, integrating ESP32/Raspberry Pi hardware with cloud dashboards to achieve real-time sensor data visualization.

Mechanical Engineering Intern | Arabic Organization for Industrialization (AOI) (Engine Factory) (Jun. 2024 - Jul. 2024)

- Mastered turbine engine overhaul procedures, including disassembly, inspection, and reassembly, through a hands-on project.
- Gained practical experience in specialized industrial processes, including nondestructive testing (NDT), chemical coating, welding technology, and heat treatment.

Technical Skills

CAD & Computational Analysis (FEA): SolidWorks, Autodesk Inventor, ANSYS, Finite Element Analysis, structural integrity validation, buckling resistance, safety factors.

Control Systems & Simulation: MATLAB, Simulink, System Modeling, Control Logic Design, Data Visualization.

IoT Technology: Data Acquisition, Sensor Fusion, Signal Processing, Condition Monitoring, MQTT, Node-RED, ThingsBoard, Cloud Dashboards.

Robotics & Autonomous Systems: ROS (Robot Operating System), Autonomous Navigation, Decision-Making.

Machine Learning & AI: Machine Learning (ML), Deep Learning, Computer Vision, Natural Language Processing (NLP), Prompt Engineering, TensorFlow, PyTorch , OpenCV, Azure ML, Predictive Maintenance, Time-Series Forecasting, Anomaly Detection, Real-Time Object Detection, Visual Inspection Systems.

Projects

Autonomous Mobile Manipulator Robot | Mechatronics & IoT Project

Technologies: PID Control, 4 DOF Robotic Arm, Computer Vision, ESP32, MQTT, Node-RED

- Designed and fabricated the complete mechanical hardware, including the autonomous mobile base structure and the 4 DOF robotic arm assembly.
- Implemented PID control for precise closed-loop base navigation alongside a Computer Vision camera module for real-time QR code detection and automated decision-making.
- Developed an IoT network using ESP microcontrollers to orchestrate seamless data flow via MQTT protocols and a Node-RED dashboard.

Predictive Maintenance System | Machine Learning & Data Engineering Project

Technologies: Machine Learning, Classification Models, API Development, MLOps, SMOTE, Data Pipelines

- Developed a binary classification model to predict machine maintenance requirements within a 7-day window based on temperature, vibration, and pressure sensor telemetry.
- Structured data preprocessing pipelines and model retraining schedules to maintain a system uptime of ≥ 99% and a target model accuracy of ≥90%.

Air-Powered Vehicle | Mechatronics & Mechanical Design Project

Technologies: MATLAB, ANSYS, ESP32, C++, SolidWorks/Inventor, TIG Welding, Pneumatics.

- Secured 4th place out of 64 competing teams by engineering a remotely controlled vehicle propelled solely by 10-bar compressed air, integrating mechanical propulsion with electronic control systems.
- Designed a lightweight chassis and 3-liter tank, performing Finite Element Analysis (FEA) via ANSYS to validate structural integrity with a Safety Factor of 2 and optimizing nozzle fluid dynamics via MATLAB simulations.
- Developed an ESP32-based PCB control unit to manage propulsion solenoid valves and braking servos to achieve a ~30cm stop distance, while overseeing all pneumatic integration and TIG welding processes.

Leadership & Extracurricular Activities

- Event Management: Coordinated logistics, scheduling, and execution for college student events to ensure seamless operations.
- Operations & Order: Maintained project workflows and kept team activities structured, organized, and on timeline.

Languages

- Arabic (Native)
- English (B2)
- Deutsch (A2)